

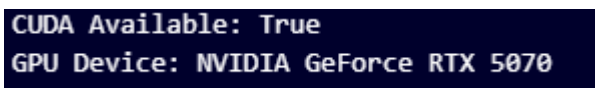
Question 1 :

A CPU is designed with a few powerful cores optimized for sequential processing, whereas a GPU is built with thousands of smaller, specialized cores designed for parallel processing. Because training deep learning models requires performing millions of simultaneous mathematical calculations, a GPU's highly parallel architecture allows it to process this massive workload drastically faster than a CPU.

Question 2 :

GPUs excel at highly parallel, repetitive mathematical calculations—often referred to as Single Instruction, Multiple Data (SIMD) operations. In the context of deep learning, these primarily include massive matrix multiplications, vector additions, and tensor operations that occur constantly during the forward pass and backpropagation.

Question 3 :



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CUDA Available: True
GPU Device: NVIDIA GeForce RTX 5070
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This output confirms that PyTorch is successfully communicating with your system's hardware. The True value means that the required CUDA drivers and libraries are installed correctly, and PyTorch recognizes your newly installed NVIDIA GeForce RTX 5070, meaning your environment is ready to offload deep learning tasks to the GPU for acceleration.

Question 4 :

Processing 10,000 images requires millions of repetitive matrix multiplications to evaluate the pixel data and update the model's weights. A GPU's highly parallel architecture allows it to process large batches of these images simultaneously, drastically reducing the total training time compared to computing them one by one on a CPU.